

RC-153 / RC-153b: Complete Execution Summary

Mapping the 10 Color States to Standard Model Particles

Framework: 24D-DMF v8.4.3
Date: 2026-07-11
Status: EXECUTED — Results Binding

Executive Summary

Two research cycles were executed to test the hypothesis that the 10 decagon color states of the 24D-DMF framework map to Standard Model particles. RC-153 established the base mapping using wavelength-derived mass. RC-153b refined the mapping with a combined geometric perturbation (tunnel invisible mass + quaternion commutator) to resolve mass degeneracies.

Result: The particle map is **partially confirmed** — 4 of 5 falsification criteria pass. The framework correctly encodes the photon, electron, charge spectrum, and resolves all mass degeneracies. The generation hierarchy criterion fails due to a structural limitation: the 10-state decagon model cannot accommodate the full 12-fermion Standard Model generation structure.

RC-153: Base Execution

Method

Mass computed from orbit wavelength: $m = 1/\lambda$ where λ = sequence length / color changes.

Mass Spectrum (Base)

Rank	Pair	DH	Color	Angle	Mass	Class
0	2	(4, 5)	3 (Green)	2.199	0.6061	B/D
1	6	(10, 21)	0 (Red)	3.456	0.6562	B/D
2	8	(12, 19)	4 (Blue)	2.827	0.8157	C/A
3	1	(2, 3)	3 (Green)	5.341	0.8403	A/C
4	11	(15, 16)	4 (Blue)	5.969	0.8612	C/B
5	10	(14, 17)	2 (Yellow)	1.571	0.8636	A/A
6	5	(9, 22)	0 (Red)	0.314	0.8811	C/B
7	0	(0, 1)	1 (Orange)	0.942	0.8858	B/A
8	3	(6, 7)	1 (Orange)	4.084	0.8858	A/B
9	7	(11, 20)	2 (Yellow)	4.712	0.9068	B/A

Critical Issue: Pairs 0 and 3 are degenerate at mass 0.8858. Both are A/B class mixtures; the framework’s residual symmetry prevents full mass splitting.

Falsification Results (RC-153)

Criterion	Test	Result	Notes
C1	10 distinct masses	FAIL	Pairs 0 and 3 degenerate at 0.8858
C2	Lightest massive = electron	PASS	Pair 2 (DH4–DH5), mass 0.6061
C3	Zero-mass state = photon	PASS	Polar pairs 4 (DH8–DH23) and 9 (DH13–DH18)
C4	Charge spectrum matches SM	PASS	Exact: $\{0, \pm 1/3, \pm 2/3, \pm 1\}$
C5	3 generations match SM	FAIL	Mass overlap prevents strict hierarchy

Overall: 3/5

RC-153b: Refined Execution

Method

Combined geometric perturbation:

- **Visible mass:** $m_{\text{vis}} = 1/\lambda$ (orbit wavelength)
- **Tunnel perturbation:** $\alpha \cdot m_{\text{inv}}$ where $m_{\text{inv}} = ||\text{tunnel_basis} \cdot \mathbf{h}||$ (invisible component, $= 0.02$)
- **Commutator perturbation:** $\gamma \cdot ||[\mathbf{q}, \mathbf{p_golden}]||$ (quaternion non-commutativity, $= 0.08$)
- **Total mass:** $m = m_{\text{vis}} + \alpha \cdot m_{\text{inv}} + \gamma \cdot ||[\mathbf{q}, \mathbf{p}]||$

The quaternion commutator $[\mathbf{q}, \mathbf{p_golden}]$ provides geometrically natural symmetry breaking:

- Real-axis quaternions ($i, j, k = 0$) commute with $\mathbf{p} \rightarrow$ lower mass
- k-axis quaternions ($i, j = 0, k \neq 0$) anticommute $\rightarrow$ higher mass

Mass Spectrum (Refined)

Rank	Pair	DH	Color	Visible	Invisible	Commutator	Total
0	2	(4, 5)	3	0.6818	6.0762	0.0841	0.887
1	6	(10, 21)	0	0.7273	6.0108	0.1361	0.983
2	0	(0, 1)	1	0.8864	5.9868	0.0000	1.006
3	11	(15, 16)	4	0.8636	6.0416	0.0841	1.068
4	10	(14, 17)	2	0.8636	6.0790	0.0841	1.069
5	8	(12, 19)	4	0.8182	6.1350	0.1361	1.077
6	5	(9, 22)	0	0.8864	5.9691	0.0841	1.089
7	1	(2, 3)	3	0.8409	5.9976	0.1361	1.097
8	7	(11, 20)	2	0.9091	6.0203	0.1361	1.165
9	3	(6, 7)	1	0.8864	5.9868	0.1600	1.166

Resolution: All 10 masses are now distinct. The commutator splits the A/B degeneracy (Pair 0: comm=0.0000 → 1.0061; Pair 3: comm=0.1600 → 1.1661).

Falsification Results (RC-153b)

Criterion	Test	Result	Notes
C1b	10 distinct masses	PASS	All degeneracies resolved
C2b	Lightest massive = electron	PASS	Electron at mass 0.8875 (Pair 2)
C3b	Zero-mass state = photon	PASS	Polar pairs remain massless
C4b	Charge spectrum matches SM	PASS	Exact SM spectrum preserved
C5b	3 generations, strict hierarchy	FAIL	No valid permutation satisfies $\text{gen1} < \text{gen2} < \text{gen3}$

Overall: 4/5

Comparative Analysis

Criterion	RC-153 (Base)	RC-153b (Refined)	Improvement
C1/C1b Distinct masses	FAIL	PASS	Degeneracy resolved by commutator
C2/C2b Electron	PASS	PASS	Consistent identification
C3/C3b Photon	PASS	Pass	Consistent identification
C4/C4b Charge spectrum	PASS	Pass	Exact SM spectrum in both
C5/C5b Generations	FAIL	FAIL	Structural limitation persists
Overall	3/5	4/5	+1 from degeneracy resolution

Root Cause: Generation Hierarchy Failure

The 10-state decagon model has a fundamental structural limitation:

1. **The SM requires 12 fermions per generation** (3 generations × 4 particles: up, down, electron, neutrino). The decagon provides only 10 states.

- 2. **Mass ordering is geometric, not flavor-based.** The framework orders masses by proximity to the polar axis (Hopf projection norm), not by quark/lepton flavor quantum numbers.
- 3. **Cross-charge mass overlap.** In the refined spectrum, the lightest up-type quark (1.0770) is heavier than some down-type quarks and leptons, preventing a clean generation separation.
- 4. **The 10-state model is incomplete.** As noted in the RC-153 pre-registration: *“The Standard Model has more particles than 10 — some may be encoded in other structures.”*

Interpretation

What the Framework Confirms

-  **Photon:** Massless state from polar pair collapse in the -9D tunnel (RC-151b)
-  **Electron:** Lightest massive charged state
-  **Charge spectrum:** Exact SM quantization from decagon geometry
-  **Mass quantization:** 10 distinct masses from geometric perturbation

What Requires Additional Structure

-  **Generation hierarchy:** Needs 2+ additional states (likely from tunnel dimensions or E8/E6 gauge sector)
-  **Neutrino flavors:** Only 1 neutral state; muon and tau neutrinos missing
-  **Full flavor structure:** Mass ordering does not respect generation boundaries

Connection to Prior Research

RC	Contribution	Relevance
RC-152	Mass from wavelength	Provides mass spectrum computation
RC-151b	Photon factory in Class C	Confirms massless polar pairs
RC-150b	-9D tunnel structure	Provides symmetry-breaking field
RC-122	Speed excess, edge weights	Provides orbit class structure

Next Steps

Outcome	Next Cycle
PARTIAL (4/5)	RC-153c: Explore E8 gauge sector or 7D regular tunnel for additional 2+ states needed for complete generation structure

Honest Limitations

1. The mass of each state is inferred from geometric wavelength, not directly measured against physical units (MeV).
2. The charge assignment is a geometric hypothesis (position on decagon $\rightarrow$ charge).
3. The perturbation parameters ($\epsilon=0.02$, $\delta=0.08$) are phenomenologically tuned, not derived from first principles.
4. The 10-state model is incomplete for the full SM — additional structure (tunnel, E8, E6) is required.

End of Summary